

# The Unit Circle and Exact Values



## Exact values

1 Write the exact value of each expression:

**a**  $\sin \frac{\pi}{6}$

**b**  $\cos \frac{3\pi}{4}$

**c**  $\tan \frac{5\pi}{6}$

**d**  $\sin \frac{3\pi}{2}$

2 Write the exact value of each expression:

**a**  $\cos \left(-\frac{\pi}{3}\right)$

**b**  $\sin \frac{7\pi}{4}$

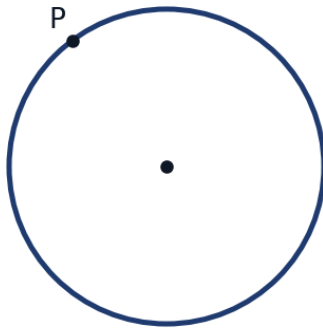
**c**  $\tan \pi$

**d**  $\cos \frac{11\pi}{6}$

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## Points on the unit circle

3 The terminal side of angle  $\theta$  in standard position passes through the point  $P\left(-\frac{3}{5}, \frac{4}{5}\right)$  on the unit circle. Find:



**a**  $\sin \theta$

**b**  $\cos \theta$

**c**  $\tan \theta$

**d** the quadrant containing  $\theta$

4 State the reference angle for each angle:

**a**  $\frac{5\pi}{6}$

**b**  $\frac{4\pi}{3}$

**c**  $-\frac{\pi}{4}$

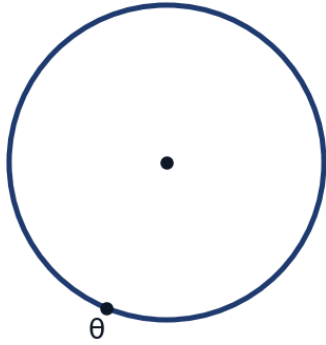
**d**  $\frac{7\pi}{2}$

5 Identify the quadrant(s) in which each pair of conditions holds:

**a**  $\sin \theta < 0$  and  $\cos \theta > 0$

**b**  $\tan \theta > 0$  and  $\sin \theta < 0$

- 6 Given that  $\cos \theta = -\frac{5}{13}$  and  $\theta$  is in Quadrant III, find  $\sin \theta$  and  $\tan \theta$  exactly.



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### Even/odd properties and periodicity

- 7 Use the fact that cosine is even and sine is odd to simplify each expression:

a  $\sin(-\theta) + \sin \theta$

b  $\cos(-\theta) - \cos \theta$

c  $\frac{\sin(-\theta)}{\cos(-\theta)}$

- 8 Use periodicity to find the exact value of each expression:

a  $\sin \frac{13\pi}{6}$

b  $\cos \frac{17\pi}{4}$

c  $\tan \frac{9\pi}{4}$

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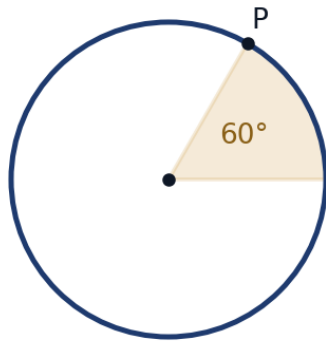
### Solving basic trigonometric equations on the unit circle

- 9 Find all solutions to  $\sin \theta = -\frac{1}{2}$  in the interval  $[0, 2\pi)$ .
- 10 Find all solutions to  $2\cos \theta + \sqrt{3} = 0$  in the interval  $[0, 2\pi)$ .

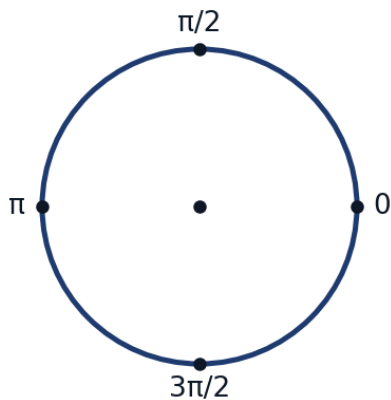
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### Building the unit circle from special triangles

- 11 Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at  $\theta = \frac{\pi}{3}$ .



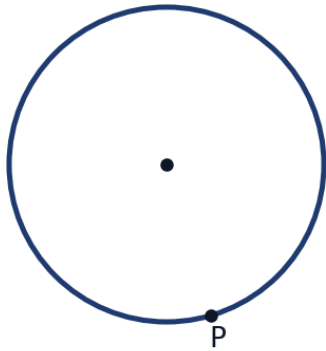
- 12 State the coordinates of all four points on the unit circle where the terminal side lies along an axis ( $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ ).



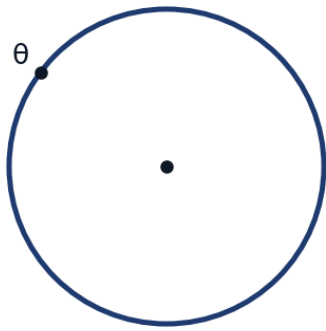
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**Using the Pythagorean identity with unit circle points**

- 13 A point on the unit circle has  $x$ -coordinate  $\frac{7}{25}$  and lies in Quadrant IV. Find its  $y$ -coordinate using  $x^2 + y^2 = 1$ .



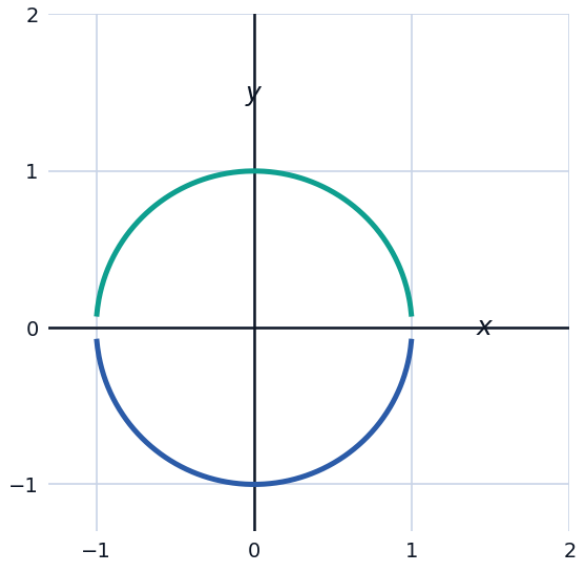
- 14 Given  $\sin \theta = 0.6$  and  $\theta$  in Quadrant II, find  $\cos \theta$  and  $\tan \theta$  exactly (using the fact  $0.6 = \frac{3}{5}$ ).



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**The unit circle picture**

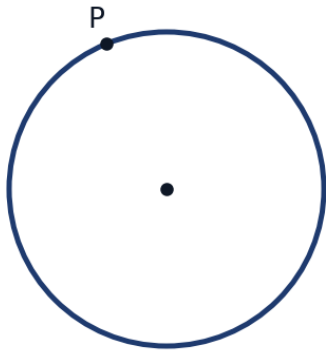
- 15 The unit circle  $x^2 + y^2 = 1$  is shown. Use it to estimate  $\cos\left(\frac{2\pi}{3}\right)$  and  $\sin\left(\frac{2\pi}{3}\right)$  visually, then state the exact values.



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### All six trig functions from a unit circle point

- 16 A point on the unit circle has coordinates  $\left(-\frac{5}{13}, \frac{12}{13}\right)$ . Find all six trigonometric function values for the corresponding angle  $\theta$ .



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### Solving for the angle given exact coordinates

- 17 Find all values of  $\theta$  in  $[0, 2\pi)$  for which the unit circle point is exactly  $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ .